CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Make a package that protects

Goal: I can protect a model product, test the package, and improve one weak point.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 5

Ask | Plan | Build | Test | Improve

You are done when

Demonstrate how your package protects and opens, and explain one improvement.

You will use

- Reclaimed small boxes
- Clean cardboard
- Paper
- String
- Masking tape
- Model products
- Rulers
- Scissors
- Shared scale if available

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Read today's goal.
- Look at the example.

2 Step 2: Tell your partner — 1 min

- Tell your partner one useful detail.

3 Step 3: Watch and try — 5 min

- Watch a model product move inside a loose package.
- Notice padding, support, and an opening.
- Watch one teacher-approved handling test.

4 Step 4: Make it work — 17 min

- Choose the reusable model product provided.
- Sketch a package that opens without damage.
- Build using a material limit shown by your teacher.

5 Step 5: Check one thing — 5 min

- Run the same gentle handling test twice.
- Check movement, damage, and ease of opening.
- Improve one weak point without adding unnecessary material.

6 Step 6: Reset together — 3 min

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

7 Step 7: Show your result — 2 min

- Demonstrate how your package protects and opens, and explain one improvement.

What to turn in

Your package and one before and after change, shown to the teacher.

No computer? Another way to do it

Cardstock package tested by hand with the same handling card.

Finished early? Next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Stay safe

Only use the teacher's handling test. Do not throw packages or test near people or equipment.

Grade targets

Grade 5: Protect one object. Show one before and after change.
Grade 6: Compare protection, material use and how easy it is to open.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.